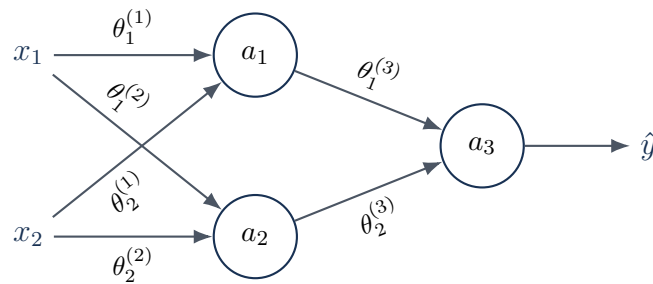


Backpropagation con regla de la cadena

Red neuronal mínima: 2 entradas, 2 neuronas ocultas y 1 salida



1. Forward

$$\begin{aligned} z_1 &= \theta_1^{(1)} x_1 + \theta_2^{(1)} x_2, & a_1 &= g(z_1) \\ z_2 &= \theta_1^{(2)} x_1 + \theta_2^{(2)} x_2, & a_2 &= g(z_2) \\ z_3 &= \theta_1^{(3)} a_1 + \theta_2^{(3)} a_2, & a_3 &= g(z_3) = \hat{y} \end{aligned}$$

$$L = L(a_3, y)$$

Idea para clase. En forward avanzamos de izquierda a derecha:

$$(x_1, x_2) \longrightarrow (z_1, z_2) \longrightarrow (a_1, a_2) \longrightarrow z_3 \longrightarrow a_3 \longrightarrow L$$

En backward recorreremos exactamente esos caminos **en sentido contrario**, multiplicando derivadas parciales.

2. Backward: capa de salida

Para $\theta_1^{(3)}$:

$$\frac{\partial L}{\partial \theta_1^{(3)}} = \frac{\partial L}{\partial a_3} \frac{\partial a_3}{\partial z_3} \frac{\partial z_3}{\partial \theta_1^{(3)}}$$

Como

$$\frac{\partial a_3}{\partial z_3} = g'(z_3), \quad \frac{\partial z_3}{\partial \theta_1^{(3)}} = a_1,$$

$$\boxed{\frac{\partial L}{\partial \theta_1^{(3)}} = \frac{\partial L}{\partial a_3} g'(z_3) a_1}$$

Para $\theta_2^{(3)}$:

$$\frac{\partial L}{\partial \theta_2^{(3)}} = \frac{\partial L}{\partial a_3} \frac{\partial a_3}{\partial z_3} \frac{\partial z_3}{\partial \theta_2^{(3)}}$$

$$\boxed{\frac{\partial L}{\partial \theta_2^{(3)}} = \frac{\partial L}{\partial a_3} g'(z_3) a_2}$$

Backward: capa oculta

Cada peso sigue su propio camino hasta la loss

Neurona oculta 1

Para $\theta_1^{(1)}$, el camino es:

$$\theta_1^{(1)} \rightarrow z_1 \rightarrow a_1 \rightarrow z_3 \rightarrow a_3 \rightarrow L$$

$$\frac{\partial L}{\partial \theta_1^{(1)}} = \frac{\partial L}{\partial a_3} \frac{\partial a_3}{\partial z_3} \frac{\partial z_3}{\partial a_1} \frac{\partial a_1}{\partial z_1} \frac{\partial z_1}{\partial \theta_1^{(1)}}$$

$$\frac{\partial L}{\partial \theta_1^{(1)}} = \frac{\partial L}{\partial a_3} g'(z_3) \theta_1^{(3)} g'(z_1) x_1$$

Para $\theta_2^{(1)}$:

$$\frac{\partial L}{\partial \theta_2^{(1)}} = \frac{\partial L}{\partial a_3} \frac{\partial a_3}{\partial z_3} \frac{\partial z_3}{\partial a_1} \frac{\partial a_1}{\partial z_1} \frac{\partial z_1}{\partial \theta_2^{(1)}}$$

$$\frac{\partial L}{\partial \theta_2^{(1)}} = \frac{\partial L}{\partial a_3} g'(z_3) \theta_1^{(3)} g'(z_1) x_2$$

Neurona oculta 2

Para $\theta_1^{(2)}$:

$$\frac{\partial L}{\partial \theta_1^{(2)}} = \frac{\partial L}{\partial a_3} \frac{\partial a_3}{\partial z_3} \frac{\partial z_3}{\partial a_2} \frac{\partial a_2}{\partial z_2} \frac{\partial z_2}{\partial \theta_1^{(2)}}$$

$$\frac{\partial L}{\partial \theta_1^{(2)}} = \frac{\partial L}{\partial a_3} g'(z_3) \theta_2^{(3)} g'(z_2) x_1$$

Para $\theta_2^{(2)}$:

$$\frac{\partial L}{\partial \theta_2^{(2)}} = \frac{\partial L}{\partial a_3} \frac{\partial a_3}{\partial z_3} \frac{\partial z_3}{\partial a_2} \frac{\partial a_2}{\partial z_2} \frac{\partial z_2}{\partial \theta_2^{(2)}}$$

$$\frac{\partial L}{\partial \theta_2^{(2)}} = \frac{\partial L}{\partial a_3} g'(z_3) \theta_2^{(3)} g'(z_2) x_2$$

Regla pedagógica: para derivar respecto a un peso, empieza en L , recorre hacia atrás el camino que conecta la loss con ese peso y multiplica todas las derivadas parciales del camino.